Solution (sketch) to IMO 2025 Shortlist

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Algebra

A1. Quadratic solitaire is a single-player game. To start the game, the player chooses two distinct nonzero integers a and b and writes the equation $x^2+ax+b=0$ on a blackboard. On each turn, if the equation currently written on the blackboard has two distinct nonzero integer solutions $x=u$ and $x=v$, then the player erases the equation and replaces it by one of the equations $x^2+ux+v=0$ and $x^2+vx+u=0$ of their choosing. Otherwise, the game ends. Determine all initial choices of a and b such that quadratic solitaire can be played forever.

Answer.$(1,-2)$ where $x^2+x-2=(x-1)(x+2)$ gives the same polynomial.

Solution. Now note that $uv=b$ so $|b|\geq \max\{|u|,|v|\}$ (as each $|u|,|v|\geq 1$) so $|b|$ is nonincreasing, hence needs to stabilize. Note that $|v|=|b|$ means $|u|=1$, hence eventually we end up with $x^2\pm x+v$ with one of $\{1,-1\}$ as root. Since $v\neq 0$, $x^2\pm x\in\{-2,0,2\}$ when $x=\pm 1$, so $|v|=2$. $x^2\pm x+2$ has no real root as discriminant is -7, hence $v=-2$. Now $x^2-x-2=(x+1)(x-2)$ has root $\{-1,2\}$, but arrangement into x^2-x+2 has no real root. Thus the only 'terminal' polynomial is x^2+x-2.

A2. The sunshine cost of a sequence a_1, a_2, …, a_{100} of integers is the largest possible value of

$$|(a_1+a_2+\cdots+a_i)-a_j|$$

as i and j vary over all integers 1, 2, … , 100.

Determine the smallest possible sunshine cost over all sequences a_1, a_2, …, a_{100} of pairwise distinct integers.

Answer. 75, or $\lceil\frac{3(n-1)}{4}\rceil$ by replacing 100 with general n that's multiple of 4.

Solution. *Lower bound*: we can't choose a_1 as the maximizer of $|a_i|$; otherwise either a_1 is the max or min and $|a_1-a_i|\geq n-1$ for some i.

Let $\max a_j-a_i=M$ satisfied by (j_0,i_0) and $\max|a_i|=C$; w.l.o.g. let $a_k=C$, $k\geq 2$. clearly can't have $|a_1|=C$. Let the sum $(a_1+\cdots+a_i)-a_j=C_{i,j}$. Now

$$C_{k,i}-C_{k-1,j}=a_k+a_j-a_i=C+M$$

so the answer is at least $\frac{C+M}{2}$. Moreover $M\geq n-1, C\geq\lceil\frac{n-1}{2}\rceil$, giving the LB.

Upper bound: we'll only show the case $n=4k$ as requested in the problem, and the desired bound is $3k$. We'll make sure all $|a_i|\leq 2k$; all $|a_1+\cdots+a_i|\leq k$. We arrange as

$$-k,2k,-2k,0,1,2k-1,-1,-(2k-1),\cdots,k-1,k+1,-(k-1),-(k+1)$$

A3. (IMO 5) Alice and Bazza are playing the inekoalaty game, a two-player game whose rules depend on a positive real number λ which is known to both players. On the nth turn of the game (starting with $n = 1$) the following happens: If n is odd, Alice chooses a nonnegative real number x_n such that

$$x_1 + x_2 + \cdots + x_n \leq \lambda n.$$

If n is even, Bazza chooses a nonnegative real number x_n such that

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leq n.$$

If a player cannot choose a suitable x_n, the game ends and the other player wins. If the game goes on forever, neither player wins. All chosen numbers are known to both players.

Determine all values of λ for which Alice has a winning strategy and all those for which Bazza has a winning strategy.

Answer. A wins when $\lambda > \frac{\sqrt{2}}{2}$; B wins when $\lambda < \frac{\sqrt{2}}{2}$.

Solution. Let $C = \frac{\sqrt{2}}{2}$.

$\lambda > C$: Coral outputs 0 at all times; Joey's outputs will make sure $\sum x_n \leq Cn$. Wait till n big enough (and even) then pick $x_{n+1} = (\lambda - C)n$ (then $\sum x_n^2 \geq (\lambda - C)n^2 > n$), Joey loses.

$\lambda < C$: Joey always tries to 'max out', i.e. $x_{2k+1}^2 + x_{2k+2}^2 = 2$. Note that $x_{2k+1} + x_{2k+2} \geq \sqrt{2}$ so Coral loses at one point.

$\lambda = C$: Coral sustains by outputting 0; Joey sustains by making sure $x_{2k+1}^2 + x_{2k+2}^2 = 2$.

A4. Let $\mathbb{Z}_{\geq 0}$ be the set of all nonnegative integers. Let $f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ be an unbounded function such that, if m and n are nonnegative integers satisfying

$$f(m+n) = \max\{f(0), f(1), \ldots, f(m+n)\},$$

then

$$f(m+n) = f(m) + f(n).$$

Prove that there exist positive integers A, B, C and D such that for all nonnegative integers n,

$$f(An + B) = Cn + D.$$

Solution. Denote peak as $k \geq 1$ satisfying $f(k) \geq f(\ell)$ for all $\ell \leq k$. Let $C = \min\{f(m) : m \geq 1\}$. There are infinitely many peaks since f is unbouned. We show the following:

$C > 0$. Indeed, if $f(m) = 0$ for $m \geq 1$ then for any peak $n \geq m$ we have $f(n) = f(n - m)$ and so f is bounded.

Exactly one m has $f(m) = c$. Let $m_1 < m_2$ be such that $f(m_1) = f(m_2) = C$. For any peak $n > m_2$, we have $f(n) = f(n - m_1) + C = f(n - m_2) + C$ and note that both $f(n - m_1), f(n - m_2)$ are $\max\{f(0), \cdots, f(n-1)\}$. Since $m_1 < m_2$ we have $f(n - m_1) = f(n - m_2) + f(m_1 - m_2)$ and $f(m_1 - m_2) = 0 < C$, which is contradiction.

If n is a peak, $f(A) = C$, and $n > A$, then $n - A$ is a peak. For any $m < n$ we have $f(n-A) = f(n) - f(A) \geq f(n) - f(m) = f(n-m)$ so $f(n-A) = \max\{f(0), \cdots, f(n-1)\}$.

The last point says the number of peaks among $\{kA+1, \cdots, (k+1)A\}$ is at most that of $\{(k-1)A+1, \cdots, kA\}$; choose such k such that this number stabilizes and let $B \in \{kA+1, \cdots, (k+1)A\}$ be a peak. Then $B, A+B, 2A+B, \cdots$ are all peaks. Therefore $f(nA+B) = nf(A) + f(B) = Cn + D$, as claimed.

A6. Let S be a set of positive integers, possibly infinite, such that no positive integer greater than 1 divides all elements of S. Determine all non-periodic infinite sequences $a_1, a_2, a_3, \ldots$ of positive integers such that, for all positive integers n,

- $a_n \leqslant |a_{n+\ell} - \ell|$ for all ℓ in S, and
- $a_n = |a_{n+\ell} - \ell|$ for at least one ℓ in S.

Answer. $a_n = n + c$ for $c \geq 0$, which obviously fulfills.

Solution. We rewrite the condition in the following form:

$$a_n = \min\{|a_{n+\ell} - \ell| : \ell \in S\} \cdots (*)$$

Lemma 1. *All bounded sequences in (*) are periodic.*

Proof. Let M be $\max\{a_n\}$, then $M < |a_{n+\ell} - \ell|$ whenever $\ell > 2M$. Denote $s_{n,k} = (a_{n+1}, \cdots, a_{n+k})$; a_n is a function of $s_{n,2M}$. Let h_n be the minimum positive integer such that $s_{n,2M} = s_{n-h_n,2M}$ (or ∞ if this number doesn't exist), then $s_{n,2M} = s_{n-h_n,2M} \to a_n = an - 2M \to s_{n-1,2M} = s_{n-h_n-1,2M}$ so $h_{n-1} \leq h_n$ and given that there are at most M^{2M} distinct possibilities of s_n, $h_n \leq M^{2M}$ for infinitely many n's, and therefore $h_n \leq M^{2M}$ for all n. Choose $h = \max\{h_n\}$, we have a_n periodic with period h. $\square$

Now among unbounded sequences, it suffices to show that $a_n = n + c$ for n sufficiently large by (*). Denote $b_n = a_n - n$ and $m = \lim_{n\to\infty} \inf\{b_n\}$. ; the second bullet point $a_n = |a_{n+\ell} - \ell|$ means either $b_{n+\ell} = b_n$ or $b_{n+\ell} = -a_n - n < b_n$; so either way $b_{n+\ell} \leq b_n$ and iteratively, we get $m \leq b_n$ for all n. On the other hand, choose a finite set S_1 such that $\gcd S_1 = 1$, n_0 such that $a_{n_0} > \max S_1$. Then $a_{n_0+\ell} \geq a_{n_0} + \ell$ for all $\ell \in S_1$ (i.e. $b_{n_0+\ell} \geq b_{n_0}$). Let p_0 be such that for all integers $p \geq p_0$, p can be written as nonnegative integer combinations of S_1 (which is valid as $\gcd S_1 = 1$, can similarly deduce $b_{n_0+p} \geq b_{n_0}$ for all $p \geq p_0$, and since $a_{n_0+p} > \max S_1$, this also gives $b_{n_1+p} \geq b_{n_1}$ for all $n_1 > n_0 + p$ (i.e. $m \geq b_n$ for all such n's). Thus combining the arguments we have $m = b_n$ for all $n > n_1$, i.e. $a_n = n + m$ for all $n > n_1$, and so the argument follows.

A7. Let $k \geqslant 3$ be a positive integer. Prove that there exists a unique k-tuplet$(x_1, x_2, \ldots, x_k)$ of positive real numbers such that $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_k$, and

$$\left\lfloor nx_1 + \frac{1}{2} \right\rfloor + \left\lfloor nx_2 + \frac{1}{2} \right\rfloor + \cdots + \left\lfloor nx_k + \frac{1}{2} \right\rfloor = n$$

for every integer n.

Answer. $x_m = \frac{2^{k-m}}{2^k - 1}$.

Solution. Call all k-tuples working if it satisfies the problem condition. We start with the following set up:

- Picking n large and using $x - 1 < \lfloor x \rfloor \leq x$, we have $\sum x_i = 1$;
- $\lfloor y + \frac{1}{2} \rfloor + \lfloor -y + \frac{1}{2} \rfloor$ is 1 if $y + \frac{1}{2}$ is integer and 0 otherwise. Considering n and $-n$ simultaneously for each $n > 0$, nx_m cannot be half-integer.
- Let S_x be integers n where $\lfloor nx + \frac{1}{2} \rfloor - \lfloor (n-1)x + \frac{1}{2} \rfloor = 1$; $S_{x_1}, \cdots, S_{x_k}$ are pairwise disjoint but union is $\mathbb{N}$.
- Any $x_1, x_2 \geq 0$ with sum 1 and nx_1 not a half-integer is a working pair.
- Denote $T_x = \min S_x \cap \mathbb{N}$; then $T_{x_1} = 1$. In addition, given that x_1 is such that nx_1 is not a half-integer, $S_{1-x_1} = \bigcup_{m=2}^k S_{x_m}$ and in particular, $T_{x_2} = T_{1-x_1} \geq 2$.
- Denote G_x the set of possible gaps between consecutive elements in S_x. We have $G_x = \mathbb{N} \cap [\lfloor \frac{1}{x} \rfloor, \lceil \frac{1}{x} \rceil]$. In addition, when $x < \frac{1}{2}$, we have $\frac{1}{2(t-1)} \leq x \leq \frac{1}{2t}$ where $t = T_x$, so $G_x \in \{2t-2, 2t-1, 2t\}$

We now consider the following steps.

Lemma 2. *$T_{x_2} = T_{1-x_1} = 2$, $G_{x_2} = \{3, 4\}$ and $G_{x_1} = \{2, 3\}$.*

Proof. Denote $T = T_{x_2}$ for simplicity. By inspecting any $t \in S_{x_3} > T_{x_2}$, and t_0, t_1 be consecutive elements in S_{x_2} such that $t_0 < t < t_1$, we have $2T \geq \max G_{x_2} \geq t_1 - t_0 \geq 2 \min G_{1-x_1} \geq 2(2T - 2)$, so $T \geq 2T - 2$ and $T = 2$. In addition, these inequalities must all hold so $\max G_{x_2} = 4, \min G_{1-x_1} = 2$. Finally, both of them must include 3, otherwise either $x_2 = \frac{1}{4}$ or $1 - x_1 = \frac{1}{2}$. These cases mean $2x_2$ and x_1 are half-integer, which is a contradiction. $\square$

Next, we note that each S_x partitions the real number line into intervals. We now consider the 'blocks' of intervals as follows: m blocks are defined by $m + 1$ points including both endpoints.

Lemma 3. *All consecutive blocks of interval length 2 in S_{1-x_1} must contain even and equal number of intervals.*

Proof. If $t \in S_{1-x_1}$ is an endpoint to an interval of length 3, and $t \notin S_{x_2}$, then S_{x_2} will have a gap of a least $2+3 \geq 5$ which contradicts $G_{x_2} = \{3, 4\}$. Conversely, for each of the intervals of size 2 in S_{1-x_1}, one of them cannot belong to x_2. This leaves each consecutive blocks to have even number of 2's, with the alternating ones (including both endpoints) belong to S_{x_2} and the other alternating ones $\cup_{i=3}^{k} S_{x_i}$.

Now, if $a \geq 2$ is the number of consecutive intervals of size 2 in S_x, i.e. for some t, $t, t+2, \cdots, t+2a \in S_x$, but $t-a, t+2a+2 \notin S_x$. Then

$$\lfloor (t+2a)x + \frac{1}{2}) \rfloor - \lfloor (t-1)x + \frac{1}{2}) \rfloor \geq a+1 \qquad \lfloor (t+2a+2)x + \frac{1}{2}) \rfloor - \lfloor (t-3)x + \frac{1}{2}) \rfloor \geq a+1$$

The first implies $(2a+1)x > a$ and $x > \frac{a}{2a+1}$; the second implies $(2a+5)x < a+2$ so $x < \frac{a+2}{2a+5}$. Thus there cannot be intervals of a blocks and b blocks simultaneously if $b \geq a+2$, proving the claim. $\square$

Consequently, if $2a$ is the consecutive block size (of interval 2) then we in fact have $1 - x_1 = \frac{2a+1}{4a+3}$, with

$$S_{1-x_1} = \{m(4a+3)+2p, p = 1, 2, \cdots, (2a+1)\} \qquad S_{x_2} = \{m(4a+3)+4p+2, p = 0, 1, \cdots, a\}$$

For convenience, denote $M = 4a + 3$. Our goal is to show that $M = 2^m - 1$. We now iteratively establish the following about S_{x_m} for $m = 3, \cdots, k$.

Lemma 4. *For all m with $m \leq k$, $U_m := \cup_{m'=m}^{k} S_{x_{m'}} = \{Mu + 2^{m-1}c, c = 1, 2, \cdots, \lfloor \frac{M}{2^{m-1}} \rfloor\}$, and in addition $U_m \cap \{1, 2, \cdots, M\}$ has odd number of elements (so $2^m \mid M+1$).*

Proof. We proceed by induction on m. For $m = 3$ the previous discussion already gave us $U_3 = \{4, 8, \cdots, M-3\} \pmod{M}$, and $4 \mid M+1$. Our discussion will first assume this form of U_m and that $2^{m-1} \mid M+1$, show that U_m has odd number of elements, $2^m \mid M+1$, and if $m < k$ the next set U_{m+1} will fulfill the first condition.

Now assume $U_m \cap \{1, 2, \cdots, M\} = \{2^{m-1}c, c = 1, 2, \cdots, \lfloor \frac{M}{2^{m-1}} \rfloor\}$, in particular $T_{x_m} = 2^{m-1}$. Since $G_{x_m} \in \{2^m - 2, 2^m - 1, 2^m\}$, this would give $S_{x_m} \cap \{1, 2, \cdots, M\} = \{2^{m-1} + 2^m c, c = 0, 2, \cdots, \lfloor \frac{M - 2^{m-1}}{2^m} \rfloor\}$. But the next element of U_m after M is $M + 2^{m-1}$, showing that

$$M + 2^{m-1} - 2^m \lfloor \frac{M - 2^{m-1}}{2^m} \rfloor \leq \max G_{x_m} \leq 2^m$$

So the remainder of $M - 2^{m-1}$ when divided by 2^m must be less than 2^{m-1}, which, combined with previous evidence, gives $M \equiv 2^m - 1 \pmod{2^m}$. This establishes that $U_m \cap \{1, 2, \cdots, M\}$ has odd number of elements (so $2^m \mid M+1$) and using $G_{x_m} \in \{2^m - 2, 2^m - 1, 2^m\}$, we iteratively have $U_{m+1} = \{Mu + 2^m c, c = 1, 2, \cdots, \lfloor \frac{M}{2^m} \rfloor\}$ whenever $m < k$. $\square$

Now to finish, we note that when $M = k$ we have $U_k = S_k$ so by the previous lemma, $U_k \cap \{1, 2, \cdots, M\}$ must have one element. Since $2^k \mid M + 1$ this is only possible when $2^k = M + 1$. With this, all the S_{x_i}'s are uniquely determined and we may show that they map exactly to the number $x_m = \frac{2^{k-m}}{2^k-1}$.

Combinatorics

C1. A line in the plane is called *sunny* if it is not parallel to any of the x–axis, the y–axis, or the line $x + y = 0$.

Let $n \geq 3$ be a given integer. Determine all nonnegative integers k such that there exist n distinct lines in the plane satisfying both of the following:

- for all positive integers a and b with $a + b \leq n + 1$, the point (a, b) lies on at least one of the lines; and
- exactly k of the n lines are sunny.

Answer. $k = 0, 1, 3$.

Solution. Let T_n be the set of allowed k's. The main claim is the following.

T_n is independent of n for all $n \geq 3$. To see why, we easily have $T_n \subseteq T_{n+1}$: just add a line (non-sunny) on the line $x + y = n$. To show $T_{n+1} \subseteq T_n$: consider the outermost triangle: union of points $x + y = n, x = 0$ and $y = 0$. We claim that one of the lines must coincide with one the three sides, which reduces to the case of n lines. Suppose otherwise, then:

- Each line not under this category has at most two intersects with the three sides in total (e.g. Menelaus') theorem. covers at most 2 lines in the outer triangle. Jointly, this gives at most $2(n + 1)$ intersections.
- On the other hand, there are exactly $3n$ points on the outer triangle, and $3n > 2(n + 1)$.

Thus it suffices to consider $n = 3$. We do three cases here.

$k = 0$ or 1 construction. $x = 0, x = 1$, and any line through $(2, 0)$, which we could choose to be sunny or not.

$k = 3$ construction. $\{(0, 2), (1, 0)\}, \{(2, 0), (0, 1)\}, \{(0, 0), (1, 1)\}$.

Why $k = 2$ fails. We split by cases: if one of the three lines are $x = 0, y = 0$ or $x + y = 2$ then one of other two lines (covering the smaller triangle) must be non-sunny (to cover 2 of out the 3 points). Otherwise, no line can cover more than 2 points (hence each line covers exactly two points). Can verify that the three lines of the $k = 3$ case are the only sunny lines through exactly two points, hence choosing 2 of them means we also need the third.

C2. There is a row of n paddocks, labelled from left to right with the integers 1 to n, where $n \geq 3$. Skippy the Kangaroo grazes in the paddock according to the following rule:

If Skippy is grazing in paddock k, then she makes a sequence of k hops from a paddock to an adjacent paddock. The first of the k hops is always to the right, unless Skippy is in paddock n, in which case it is to the left. Each of the following $k - 1$ hops is in the same direction as the previous hop, unless Skippy is in paddock 1 or n. Skippy grazes in the paddock that she is in after the k^{th} hop.

For example, if $n = 8$ and Skippy is grazing in paddock 3, the next 4 paddocks she will graze in are 6, 4, 8, 2, in this order. Skippy continues grazing in this way indefinitely. A paddock is overgrazed if Skippy eventually grazes in it, regardless of which paddock she starts in. Determine all $n \geq 3$ for which there exists exactly one overgrazed paddock.

Answer. $n = 3 \cdot 2^k$.

Solution. Denote $f(k)$ as the next grazing grid of Skippy when she is grazing on k. If k is overgrazed, so is $f(k), f(f(k)), \cdots$. Therefore if k is unique overgrazer, $k = f(k)$. We now note the following rule:

$$f(k) = \begin{cases} 2k & k \leq \frac{n}{2} \\ 2(n-k) & \frac{n}{2} < k \leq n-1 \\ 2 & k = 2 \end{cases}$$

We start with these observations:

- If $k = f(k)$, then $k = \frac{2n}{3}$, or $3 \mid n$.
- If $k \neq n$, and $\nu_2(k) \leq \nu_2(n)$, then $\nu_2(f(k)) > \nu_2(k)$.
- If $k \neq n$, $p > 1$ odd, $p \mid n$, then $p \mid k$ if and only if $p \mid f(k)$.

In particular, the first bullet point allows us to disregard powers of 2. The next we will eliminate is where n is divisible by pq where $p, q > 1$ are odd. The last bullet point suggests that Skippy will only cycle among grids not divisible by p when she starts at 1. However, if she starts at p, the subsequent grids she stays at cannot be divisible by pq, and therefore cannot land at n at all. On the other hand, this also means all the grids it stays are divisible by p. Thus the two paths has two terminal cycles that are disjoint, showing that there is no overgrazed pollock for tis n.

We are left with the case $n = p \cdot 2^k$, where p is prime. Given the requirement $3 \mid n$, we have $n = 3 \cdot 2^k$ for $k \geq 0$. We now claim 2^{k+1} is the only overgrazed pollock: in fact, all paths would terminate here. When $m = 2^{k+1}, f(m) = m$. When $m = n$, $f(m) = 2$ and eventually leads to 2^{k+1}. In all other cases $\nu_2(f(m)) \geq \nu_2(m) + 1$, so ν_2 increases as Skippy goes (possibly hitting $m = n$ but then goes back to the cycle to 2^{k+1}.)

C3. There are 2025 white balls and 2025 black balls arranged in a row. A balanced segment is a contiguous nonempty segment of balls that contains the same number of white balls as black balls. A balanced removal is the operation of selecting a balanced segment S, removing all the balls in S, and shifting left the balls that were to the right of S so that the remaining balls form a contiguous row.

Determine the smallest positive integer N satisfying the following property: for every configuration of balls and for every integer k satisfying $0 \leq k \leq 2025$, there exists a sequence of at most N balanced removals after which there are exactly $2k$ remaining balls.

Answer. $N = 2$.

Solution. We generalize to n white and n black balls, $n \geq 4$. To see why $N = 1$ fails, consider any configuration when first 2 and last 2 balls are all white; then $k = 1$ fails here.

To see why $N = 2$ always works, we consider the notion of 'prefix' as the subset of first m balls for any m; call this prefix strict if $m < 2k$; denote suffix similarly (as consecutive m balls). We start with the following observation:

$N = 1$ works when any strict prefix has more white than black balls (or vice versa). W.l.o.g. we only consider the first case, and the task of removing $2(n-k)$ consecutive balls. Let m_g be the difference between white and black balls among balls $g+1, \cdots, g+2(n-k)$ for $g = 1, 2, \cdots, 2k$. Then $m_{2k} < 0 < m_0$, m_i's are all even, and morever $|m_g - m_{g+1}| \in \{0, 2\}$. Hence the conclusion follows by discrete continuous value theorem.

To finish, consider the smallest prefix P of at least $2k$ balls and is balanced; the first move would be remove the suffix of the whole sequence, with P left. Let Q be the smallest suffix of P that is balanced, and let $m = |P| - 2k > |Q|$ be the number of balls we need to remove. By minimality, the condition of our previous claim applies, hence we may remove m balls from this suffix Q in one step.

C4. Bluey is placing lollies on a 1000×1000 grid. Initially, every cell of the grid is empty. At each step, Bluey chooses a row or column in which the total number of lollies is a multiple of 3, chooses an empty cell in this row or column, and places a lolly in it. If there is no such row or column, Bluey stops.

Determine the maximum number of lollies that Bluey can place.

Answer. 3995 (or $4n - 5$ for all $n \geq 5$).

Solution. *Bound:* let the 'score' of each row and column be 0, 1, 2 when the number of lollies is congruent to 0, 2, 1, respectively. Denote total score as total row score and total column score across all n rows and columns (i.e. range 0 to $4n$). Consider each of the lolly operation (modulo 3) :

$$(0,0) \to (1,1), (0,1) \to (1,2), (0,2) \to (1,0)$$

Call this type 0, 1, 2. Corresponding score update is $+4, +1, +1$, respectively, and the first update must be type 0 (hence $+4$ points). Now the last update:

- If it's type 0, max total number of steps is $4n - 8 + 2 = 4n - 6$;
- If type 1 or 2, then there is row (and column) where the score is ≤ 2. The total row score and total column score must align modulo 3 (hence $\leq 2n - 1$ each); total score $\leq 4n - 2$ so including the first 4-point update total number of steps is $\leq 4n - 5$.

Construction: first do

$$(1,1), (i, i+1), (i+1, i), i = 1, 2, \cdots, n-2$$

Now consider permutation π such that $\pi(1) = n$, $\pi(n) = n - 1$ and $|\pi(i) - \pi(i+1)| \neq 1$ for all $i \geq 1$. E.g. for n even we may do

$$1, 3, \cdots, n-3, n, n-2, \cdots, 2, n-1$$

for n odd:

$$1, 3, \cdots, n, 2, 4, \cdots, n-1$$

Then row/column $\pi(1)$ has 0 lolly, $\pi(n)$ has 1, and $\pi(i)$ has 2 for $i = 2, \cdots, n-1$. Then do

$$(\pi(i), \pi(i+1)), (\pi(i+1), \pi(i)), i = 1, 2, \cdots, n-1$$

which gives total of $4n - 5$ operations.

C5. Let $a_1, a_2, \ldots, a_n$ be a permutation of $1, 2, \ldots, n$. A pair (i, j) with $i < j$ is called an inversion if $a_i > a_j$. An inversion is called odd if $i - j$ is odd.

(a) Prove that the total number of inversions is at most 5 times the number of odd inversions.

(b) There is a permutation such that the total number of inversions is more than 4.99 times the number of odd inversions.

Solution.

(a) Define 'even inversion' similarly. Note that for any inversion (i, j) with $i < j$, any k with $i < k < j$, one (or both) of (i, k) and (k, j) must also be an inversion.

We construct a bipartite graph with one side even inversion, and the other side odd inversion, in the following manner: for every even inversion (i, j) with $j - i = 2d > 0$, if d is odd, we connect an edge of weight 1 to either $(i, i + d)$ or $(i + d, j)$, whichever is an inversion; if d is even, we consider one of $(i, i+d-1), (i+d-1, j)$ that is an inversion, and also one of $(i, i+d+1), (i+d+1, j)$ that is an inversion, and connect (i, j) to each of the chosen inversions with weight $\frac{1}{2}$. Now note that each even inversion is incident edges of total weight 1, thus the total weight of edges is $A - B$. Meanwhile for each odd inversion (i, j) with $j = i + d$, the possible neighbours are:

$$(i, j + d), (i - d, j), (i, j + d + 1), (i - d - 1, j), (i, j + d - 1), (i - d + 1, j)$$

The first two (if chosen) have weights 1 each and the remaining four have weight $\frac{1}{2}$ each, so each of odd inversions are incident to edges of total weight at most $2 + 4 \cdot 12 = 4$. It follows that we have $A - B \leq 4B$ or $A \leq 5B$.

(b) Let $k, m > 1$ and $n = 2km$. We consider m blocks, each $2k$ elements. Then organize the n numbers as follows:

- We consider $2m$ classes of k numbers each, if $i < j$ then elements in class i is strictly smaller than elements in class j (w.l.o.g. can treat the k elements of class j as $jk, jk + 1, \cdots, (j + 1)k - 1$).
- Each block gets two classes; the only inversion at block level is that for two adjacent blocks i and $i + 1$, bigger class at block i is bigger than smaller class at block $i + 1$.
- Within each block i of classes a, b $(a < b)$, the numbers are presented in alternating and decreasing order; if i is odd then odd indices within this block take class a and even indices take class b. Else the opposite.

E.g. when $k = m = 4$, the classes will take the form

$$(0, 2), (1, 4), (3, 6), (5, 7)$$

The first block will have number (denoted in (class, number) for brevity so (a, b) means $ak + b$)

$$(0, 3), (2, 3), (0, 2), (2, 2), (0, 1), (2, 1), (0, 0), (2, 0)$$

and the second block

$$(4, 3), (1, 3), (4, 2), (1, 2), (4, 1), (1, 1), (4, 0), (1, 0)$$

We now claim that when m is even, $B = \frac{k^2 m}{2}$, and $A - B = k(k - 1)m + k^2(m - 1)$.

Now, inversion involving pair of different blocks could only happen for adjacent blocks: bigger class at block i vs smaller class at block $i+1$ $(i = 1, 2, \cdots, m-1)$; by construction, such inversions must be even and contributes to k^2 each, thus contributing $k^2(m - 1)$ to $A - B$.

Within the same block, the two sets of k numbers of the same class are of even inversion, thereby contributing to $2\binom{k}{2} = k(k - 1)$ inversions. Taken together, this contributes another $k(k - 1)m$ to $A - B$. Finally, for the same block i but different classes, around half are odd inversions: $\binom{k}{2}$ for i odd but $\binom{k+1}{2}$ for i even. This gives a total of $\frac{k^2 m}{2}$ odd inversions. Therefore we have

$$\frac{A}{B} = 1 + \frac{k(k - 1)m + k^2(m - 1)}{k^2 m/2} = 1 + \frac{2(k - 1)}{k} + \frac{2(m - 1)}{m} = 5 - \frac{2}{k} - \frac{2}{m}$$

Taking $k = m = 1000$ solves the problem.

C7. Let P be a regular $100-$gon. An Aussie triangulation is formed by dividing P into a set T of non-overlapping triangles such that:

- there are exactly 2025 different points, including the 100 vertices of P that are a vertex of at least one triangle in T, and
- no three of these 2025 points are collinear.

A quokkalateral is a convex quadrilateral that consists of two triangles in T sharing a common side. Two quokkalaterals may share a triangle in T. Determine the minimum number of quokkalateral among all Aussie triangulation.

Geometry

G1.

G4. (IMO 2) Let Ω and Γ be circles with centres M and N, respectively, such that the radius of Ω is less than the radius of Γ. Suppose Ω and Γ intersect at two distinct points A and B. Line MN intersects Ω at C and Γ at D, so that C, M, N, D lie on MN in that order. Let P be the circumcentre of triangle ACD. Line AP meets Ω again at $E \neq A$ and meets Γ again at $F \neq A$. Let H be the orthocentre of triangle PMN.

Prove that the line through H parallel to AP is tangent to the circumcircle of triangle BEF.

Number Theory

N1. Let n and k be positive integers such that $n < k < 2n$. Suppose $a_1, a_2, \ldots, a_k$ are positive integers such that $2^{a_1} + 2^{a_2} + \cdots + 2^{a_k}$ is divisible by $2^n - 1$.

Show that at least three of $a_1, a_2, \ldots, a_k$ have the same remainder when divided by n.

N2. In an $n \times n$ board, for each $1 \leq i \leq n$ and $1 \leq j \leq n$, the cell in the ith row from the bottom and jth column from the left contains $\gcd(i, j)$ leaves. A koala travels from the bottom left corner to the top right corner. At each step, the koala moves up or to the right by a single cell. The koala eats the leaves in each cell it visits, including the bottom left and top right cells.

Determine the maximum number of leaves that the koala can eat.

N3. A proper divisor of a positive integer N is a positive divisor of N other than N itself.

The infinite sequence $a_1, a_2, \cdots$ consists of positive integers, each of which has at least three proper divisors. For each $n \geq 1$, the integer a_{n+1} is the sum of the three largest proper divisors of a_n.

Determine all possible values of a_1.

N4. Find all functions $f : \mathbb{N} \to \mathbb{Z}$ such that for every positive integers n and d such that $d \mid n$, there exits a positive integer $e \mid n$ such that $n \mid d + f(e)$.

N5. The sequence of triples of positive integers $(a_0, b_0, c_0), (a_1, b_1, c_1), \ldots, (a_{2025}, b_{2025}, c_{2025})$ satisfies

$$(a_{i+1}, b_{i+1}, c_{i+1}) = \big(a_i + \gcd(b_i, c_i), b_i + \gcd(c_i, a_i), c_i + \gcd(a_i, b_i)\big)$$

for $0 \leqslant i \leqslant 2024$. Determine the smallest possible value of a_{2025}.

N6. For positive integers n and k, let $f_k(n)$ denote the remainder when n is divided by $2^k - 1$. A positive integer n is grouseif

$$f_1(n) \leq f_2(n) \leq f_3(n) \leq f_4(n) \leq \dots$$

(a)Prove that there are infinitely many grouse numbers.

(b) Prove that there exists a positive integer M such that there are at most $M^{1/2025}$ grouse numbers less than M.

N7. Let $\mathbb{N}$ denote the set of positive integers. A function $f\colon \mathbb{N} \to \mathbb{N}$ is said to be bonza if

$$f(a) \text{ divides } b^a - f(b)^{f(a)}$$

for all positive integers a and b.

Determine the smallest real constant c such that $f(n) \leqslant cn$ for all bonza functions f and all positive integers n.

N8. Prove that there are finitely many prime numbers p such that

- $p - 8$ is a perfect square, and
- for all odd positive integers $k < \sqrt{p}$, the number $p - k^2$ has at most two distinct prime factors.